CHAPTER 37

Plane Geometry

SECTION I LINES AND ANGLES

Line A geometrical straight line is a set of points that extends endlessly in both the directions.

Axiom-1 A line contains infinitely many points.

Axiom-2 Through a given point, infinitely many lines pass.

Axiom-3 Given two distinct points A and B, there is one and only one line that contains both the points.

Parallel Lines If two lines have no point in common, they are said to be *parallel lines*

Intersecting Lines If two lines have a point in common, they are said to be *intersecting lines*. Two lines can intersect at the most at one point.

Line Segment and Ray A part (or portion) of a line with two end points is called a *line segment* and a part of a line with one end point is called a *ray*. A line segment $\overline{AB}$ and its length is denoted as AB. Ray AB (i.e., A towards B) is denoted as $\vec{AB}$ and ray BA (i.e., B towards A) is denoted as $\vec{BA}$.

Collinear Points Three or more than three points are said to be *collinear* if there is a line which contains them all.

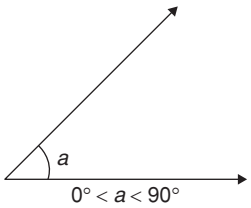
Concurrent Lines Three or more than three lines are said to be *concurrent* if there is a point which lies on all of them.

Angle An *angle* is a figure formed by two rays with a common initial point. The two rays forming an angle are called *arms* of the angle and the common initial point is called *vertex* of the angle.

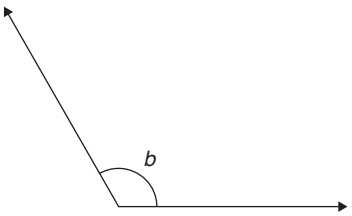
Types of Angles

An angle is said to be:

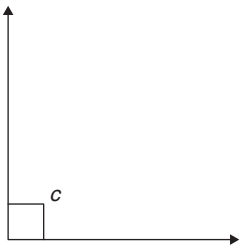
(i) *Acute*, if $a < 90^\circ$.



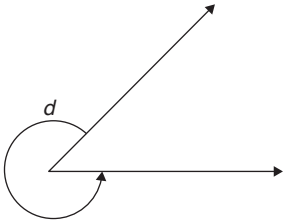
(ii) *Obtuse*, if $90^\circ < b < 180^\circ$.



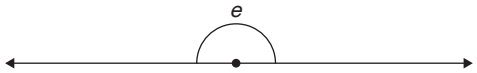
(iii) *Right angle*, if $c = 90^\circ$



(iv) *Reflex angle*, if $180^\circ < d < 360^\circ$



(v) *Straight angle*, if $e = 180^\circ$



(vi) *Complete angle*: An angle whose measure is 360º, is called a *complete angle*.

Complementary Angles Two angles, the sum of whose measures is 90º, are called *complementary angles,* e.g. 50º and 40º is a pair of complementary angles.

Supplementary Angles Two angles, the sum of whose measures is 180º, are called *supplementary angles*, e.g. 72º and 108º is a pair of supplementary angles.

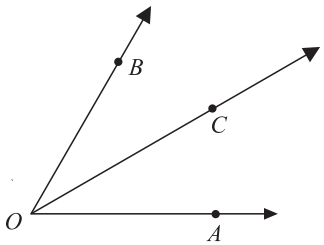
Adjacent Angles Two angles are called *adjacent angles* if

(i) they have the same vertex.

(ii) they have a common arm.

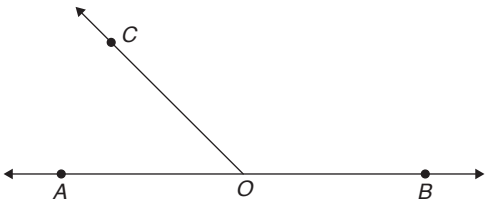
(iii) uncommon arms are on either side of the common arm.

E.g. $\angle AOC$ and $\angle BOC$ are adjacent angles.



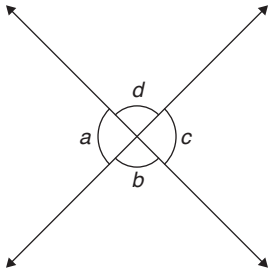
Linear Pair: Two adjacent angles are said to form a linear pair of angles if their non-common arms are two opposite rays.

E.g. $\angle AOC$ and $\angle BOC$ form a linear pair.



Linear Pair Axiom: If a ray stands on a line, then the sum of the two adjacent angles so formed is 180º. Conversely, if the sum of two adjacent angles is 180º; then the non-common arms of the angles are two opposite rays.

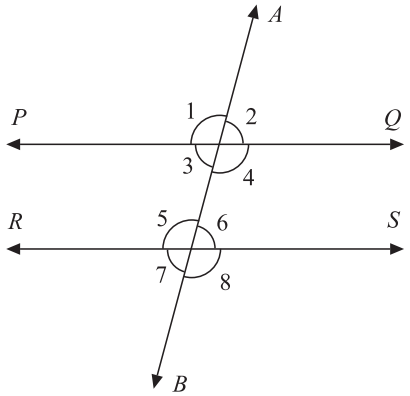
Vertically Opposite Angles: When two lines intersect, four angles are formed. The angles opposite to each other are called *vertically opposite angles*.



a and c are vertically opposite angles, $\angle a = \angle c$.

b and d are vertically opposite angles, $\angle b = \angle d$.

Angles made by a transversal* with two parallel lines Suppose $PQ \parallel RS$ and a transversal AB cuts them, then



(a) Pair of corresponding angles are
(1 and $\angle 5$), ($\angle 2$ and $\angle 6$),
($\angle 4$ and $\angle 8$) and ($\angle 3$ and $\angle 7$)

(b) Pair of alternate angles are
($\angle 3$ and $\angle 6$) and ($\angle 4$ and $\angle 5$)

(c) Pair of interior angles (consecutive interior angles or cointerior angles) on the same side of the transversal are
($\angle 3$ and $\angle 5$) and ($\angle 4$ and $\angle 6$)

KEY RESULTS TO REMEMBER

If two parallel lines are intersected by a transversal, then

(i) each pair of corresponding angles are equal.

(ii) each pair of alternate angles are equal.

(iii) interior angles on the same side of the transversal are supplementary.

Triangle A plane figure bounded by three lines in a plane is called a *triangle*.

*A line which intersects two or more lines at distinct points is called a transversal of the given lines.

SECTION 2 TRIANGLES

Types of Triangles (On the basis of sides)

Scalene triangle A triangle two of whose sides are equal is called a *scalene triangle*.

Isosceles triangle A triangle two of whose sides are equal in length is called an *isosceles triangle*.

Equilateral triangle A triangle all of whose sides are equal is called an *equilateral triangle*.

Types of Triangles (On the basis of angles)

Acute triangle A triangle, each of whose angle is acute, is called an *acute triangle* or *acute-angled triangle*.

Right triangle A triangle with one right angle is called a *right triangle* or a *right-angled triangle*.

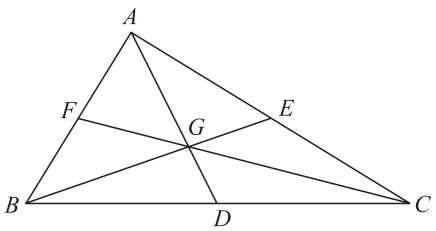
Obtuse triangle A triangle with one angle an obtuse angle, is known as *obtuse triangle* or *obtuse-angled triangle*.

Some Important Terms Related to a Triangle

1. **Median** The median of a triangle corresponding to any side is the line segment joining the midpoint of that side with the opposite vertex.

   In the figure given below, *AD, BE* and *CF* are the medians.

   The medians of a triangle are concurrent i.e., they intersect each other at the same point.

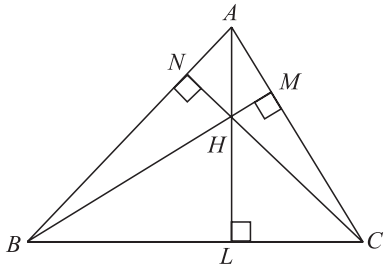


2. **Centroid** The point of intersection of all the three medians of a triangle is called its *centroid.*

   In the above figure G is the centroid of ΔABC.

   Note: The centroid divides a median in the ratio 2:1.

3. **Altitudes** The *altitude* of a triangle corresponding to any side is the length of perpendicular drawn from the opposite vertex to that side.



In the figure given above, *AL*, *BM* and *CN* are the altitudes.

Notes

The altitudes of a triangle are concurrent.

4. **Orthocentre** The point of intersection of all the three altitudes of a triangle is called its *orthocentre*. In the figure given above H is the orthocentre of ΔABC.

Notes

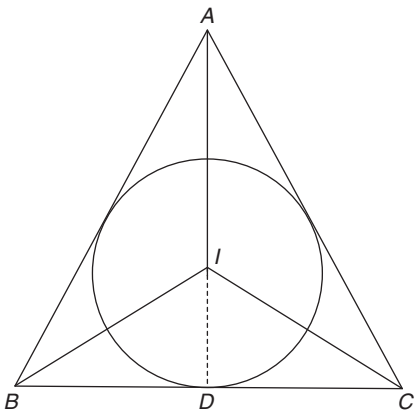
5. **Incentre of a triangle** The point of intersection of the internal bisectors of the angles of a triangle is called its *incentre*.

   In the figure given below, the internal bisectors of the angles of ΔABC intersect at I.

   $\therefore$ I is the Incentre of ΔABC.

   Let, $ID \perp BC$

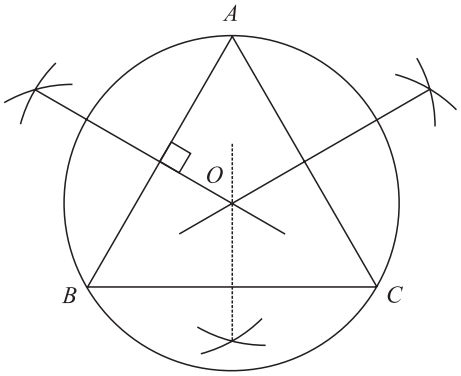
   Then, a circle with centre I and radius ID is called the *incircle* of ΔABC.



Notes

6. Circumcentre of a triangle The point of intersection of the perpendicular bisectors of the sides of a triangle is called its *circumcentre*.

In the figure given below, the right bisectors of the sides of ΔABC intersect at O.



$\therefore$ O is the *circumcentre* of ΔABC with O as centre and radius equal to $OA = OB = OC$. We draw a circle passing through the vertices of the given Δ. This circle is called the *circumcircle* of ΔABC.

Notes

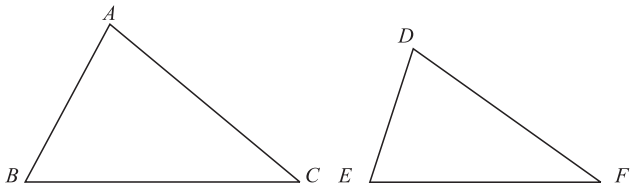
Note: The circumcentre of a triangle is *equidistant* from its *vertices*.

Congruent triangles

Two triangles are *congruent* if and only if one of them can be superposed on the other, so as to cover it exactly.

Thus, congruent triangles are exactly identical

For example, If $\Delta ABC \cong \Delta DEF$ then we have



$\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$;

and $AB = DE$, $BC = EF$ and $AC = DF$.

Similar Triangles

Congruent figures Two geometric figures having the same shape and size are known as *congruent figures*.

Similar figures Two figures (plane or solid) are said to be *similar* if they have the same shape irrespective of their sizes.

Note: Two similar figures may not be congruent as their size may be different.

For examples,

1. Any two lines segments are similar.
2. Any two equilateral triangles are similar.
3. Any two squares are similar.
4. Any two circles are similar.
5. Any two rectangles are similar.

Similar triangles Two triangles are similar if

(a) their corresponding angles are equal.

(b) their corresponding sides are proportional.

KEY RESULTS TO REMEMBER

1. The sum of all the angles round a point is equal to 360º.
2. Two lines parallel to the same line are parallel to each other.
3. The sum of three angles of a triangle is 180º.
4. If a side of a triangle is produced, the exterior angle so formed is equal to the sum of the two interior opposite angles. (Exterior Angle Theorem)
5. If two sides of a triangle are unequal, the longer side has greater angle opposite to it.
6. In a triangle, the greater angle has the longer side opposite to it.
7. The sum of any two sides of a triangle is greater than the third side.
8. If a, b, c denote the sides of a triangle then
   (i) If $c^2 < a^2 + b^2$, triangle is acute angled.
   (ii) If $c^2 = a^2 + b^2$, triangle is right angled.
   (iii) If $c^2 > a^2 + b^2$, triangle is obtuse angled.
9. Two triangles are congruent if:
   (i) Any two sides and the included angle of one triangle are equal to any two sides and the included angle of the other triangle.
   (SAS congruence theorem)
   (ii) Two angles and the included side of one triangle are equal to the corresponding two angles and the included side of the other triangle.
   (ASA congruence theorem)
   (iii) The three sides of one triangle are equal to the corresponding three sides of the other triangle.
   (SSS congruence theorem)

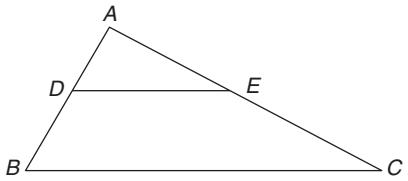
Notes

Two right triangles are congruent if the hypotenuse and one side of one triangle are respectively equal to the hypotenuse and the corresponding side of the other triangle.
(RHS Congruence theorem)

10. The line segments joining the mid-points of any two sides of a triangle is parallel to the third side and equal to half of it.

11. Basic Proportionality Theorem If a line is drawn parallel to one side of a triangle, to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.

In the figure given below, In a ΔABC



If $DE \parallel BC$

Then, $\frac{AD}{DB} = \frac{AE}{EC}$

Illustration 1: In the figure given above, D and E are the points on the AB and AC respectively such that $DE \parallel BC$. If $AD = 8$ cm, $AB = 12$ cm and $AE = 12$ cm. Find CE

Solution: In ΔABC, $DE \parallel BC$

$$\frac{AD}{DB} = \frac{AE}{EC} \text{ (Basic Proportionality Theorem)}$$

$$\Rightarrow \quad \frac{8}{12-8} = \frac{12}{EC}$$

$$\Rightarrow \quad \frac{8}{4} = \frac{12}{EC}$$

or $EC = 6$ cm

12. If a line divides any two sides of a triangle in the same ratio, the line is parallel to the third side.

Explanation In the above figure (given in point 11). In ΔABC

if $\frac{AD}{DB} = \frac{AE}{EC}$, then $DE \parallel BC$.

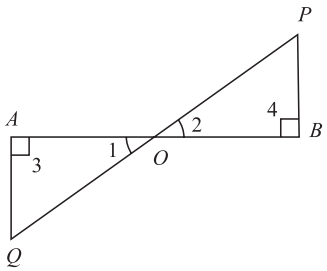
Similarity Theorems

13. AAA Similarity If in two triangles, corresponding angles are equal, then the triangles are similar.

Corollary (AA-similarity): If two angles of one triangle are respectively equal to two angles of another triangle then the two triangles are similar.

Illustration 2: In the figure given below, QA and PB are perpendiculars to AB. If $AO = 15$ cm, $BO = 9$ cm, $PB = 12$ cm, find AQ.

Solution:



In Δs AOQ and BOP

$\angle 1 = \angle 2$ [vertically opposite angles]

$\angle 3 = \angle 4$ [each 90º]

$\therefore \quad \Delta AOQ \sim \Delta BOP$ [AA Similarity Criterion]

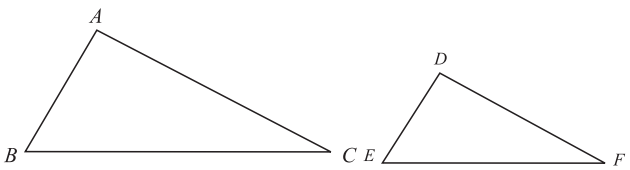
$$\therefore \quad \frac{AO}{BO} = \frac{AQ}{BP} \text{ (corresponding sides of } \sim\Delta\text{s)}$$

$$\text{or} \quad \frac{15}{9} = \frac{AQ}{12}$$

$$\text{or} \quad \frac{5}{1} = \frac{AQ}{4} \Rightarrow AQ = 20 \text{ cm.}$$

14. SSS-Similarity If the corresponding sides of two triangles are proportional then they are similar.

Explanation: In Δs ABC and DEF,

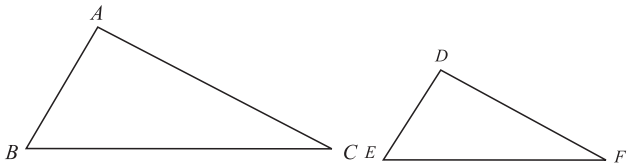


if $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$

Then $\Delta ABC \sim \Delta DEF$ [SSS Similarity]

15. SAS-Similarity If in two triangles, one pair of corresponding sides are proportional and the included angles are equal, then the two triangles are similar.

Explanation In Δs *ABC* and *DEF,*



if $\angle A = \angle D$ and $\frac{AB}{DE} = \frac{AC}{DF}$

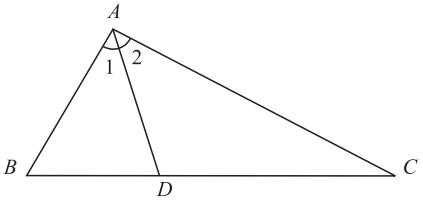
or $\angle B = \angle E$ and $\frac{AB}{DE} = \frac{BC}{EF}$

or $\angle C = \angle F$ and $\frac{AC}{DF} = \frac{BC}{EF}$,

then $\Delta ABC \sim \Delta DEF$ [SAS-Similarity]

16. Internal Bisector Property The internal bisector of an angle of a triangle divides the opposite side in the ratio of the sides containing the angle.

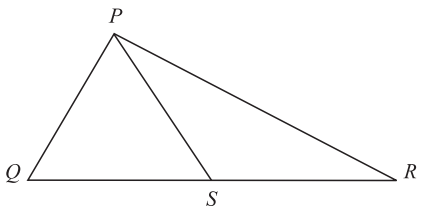
Explanation In ΔABC, if $\angle 1 = \angle 2$



Then $\frac{AB}{AC} = \frac{BD}{CD}$

17. If a line segment drawn from the vertex of an angle of a triangle to its opposite side divides it in the ratio of the sides containing the angle, then the line segment bisects the angle.

Illustration 3: In ΔPQR, $PQ = 6$ cm, $PR = 8$ cm,
Solution: $QS = 1.5$ cm, $RS = 2$ cm



$\therefore \quad \frac{PQ}{PR} = \frac{6}{8} = \frac{3}{4}$ and $\frac{QS}{RS} = \frac{1.5}{2} = \frac{3}{4}$

Thus, $\frac{PQ}{PR} = \frac{QS}{RS}$

$\therefore$ *PS* is the bisector of $\angle P$.

18. Pythagoras Theorem In a right triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

Explanation In a right ΔABC, right angled at *B*

$$AC^2 = AB^2 + BC^2$$

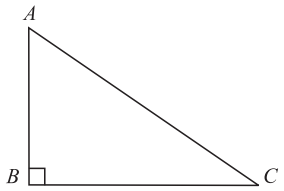
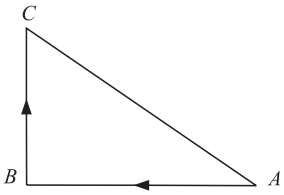


Illustration 4: A man goes 15 m west and then 8 m due north. How far is he from the starting point.

Solution: Let, the initial position of the man be *A*.



Let, $AB = 15$ m and $BC = 8$ m

$\therefore \quad AC^2 = AB^2 + BC^2$ (Pythagoras Theorem)

$= (15)^2 + (8)^2$

$= 225 + 64$

$= 289$

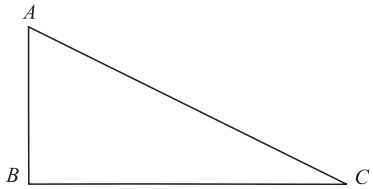
$AC = \sqrt{289}$

$= 17$ m

Hence, the man is 17 m away from the starting point.

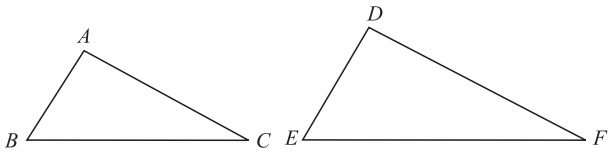
19. Converse of Pythagoras Theorem. In a triangle, if the square of one side is equal to the sum of the squares of the other two sides, then the angle opposite the first side is a right angle.

Explanation In a ΔABC if $AB^2 + BC^2 = AC^2$
Then, $\angle ABC = 90^\circ$



20. Area Theorem The ratio of the areas of two similar Ds is equal to the ratio of the squares of any two corresponding sides

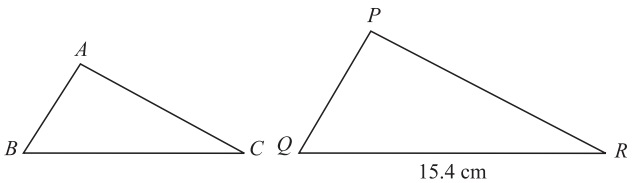
Explanation If $\Delta ABC \sim \Delta DEF$,



$$\text{then } \frac{ar(\Delta ABC)}{ar(\Delta DEF)} = \frac{AB^2}{DE^2} = \frac{AC^2}{DF^2} = \frac{BC^2}{EF^2}$$

Illustration 5: The areas of two similar Δs ABC and PQR are 64 cm^2 and 121 cm^2, respectively. If QR = 15.4 cm, find BC.

Solution: Since $\Delta ABC \sim \Delta PQR$



$$\therefore \quad \frac{ar(\Delta ABC)}{ar(\Delta PQR)} = \frac{BC^2}{QR^2} \text{ (Area Theorem)}$$

$$\text{i.e., } \frac{64}{121} = \frac{BC^2}{(15.4)^2} \Rightarrow \frac{8}{11} = \frac{BC}{15.4}$$

$\therefore \quad BC = 11.2$ cm

21. The ratio of the areas of two similar triangles is equal to the

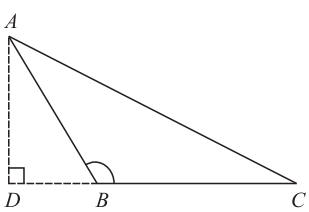
(i) ratio of the squares of the corresponding medians
(ii) ratio of the squares of the corresponding altitudes
(iii) ratio of the squares of the corresponding angle bisector segments

22. If two similar triangles have equal areas, then the Ds are congruent.

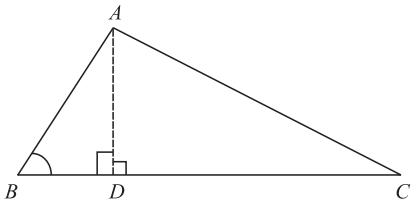
23. In two similar triangles, the ratio of two corresponding sides is same as the ratio of their perimeters.

24. Obtuse Angle Property in a ΔABC, if $\angle B$ is obtuse then $AC^2 = AB^2 + BC^2 + 2\ BC \times BD$

where $AD \perp BC$

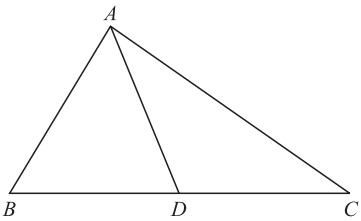


25. Acute Angle Property In a ΔABC, if $\angle C$ is acute, then $AB^2 = AC^2 + BC^2 - 2BC \times CD$ where $AD \perp BC$



26. Apollonius Theorem The sum of the squares on any two sides of a triangle is equal to the sum of twice the square of the median, which bisects the third side and half the square of the third side.

Explanation In the given ΔABC,



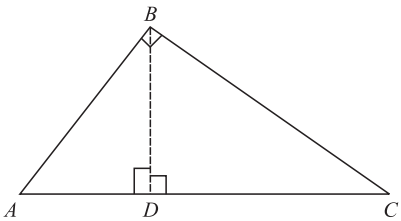
$$AB^2 + AC^2 = 2AD^2 + \frac{1}{2}BC^2$$

or $\quad AB^2 + AC^2 = 2[AD^2 + BD^2]$

27. If a perpendicular is drawn from the vertex of the right angle of a right triangle to the hypotenuse, the triangles on each side of the perpendicular are similar to the whole triangle and to each other. Also the square of the perpendicular is equal to the product of the lengths of the two parts of the hypotenuse.

Explanation In the figure given below,

ABC is a right triangle, right angled at B and $BD \perp AC$, then



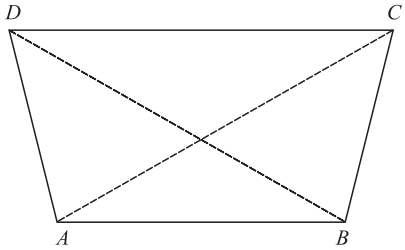
(i) $\Delta ADB \sim \Delta ABC$ (AA Similarity)

(ii) $\Delta BDC \sim \Delta ABC$ (AA Similarity)

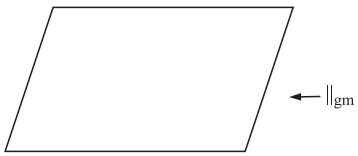
(iii) $\Delta ADB \sim \Delta BDC$ also $BD^2 = AD \times CD$

SECTION 3 QUADRILATERALS AND PARALLELOGRAMS

Quadrilateral A plane figure bounded by four line segments *AB*, *BC*, *CD* and *DA* is called a *quadrilateral*, written as quad. *ABCD* or $\angle ABCD$.

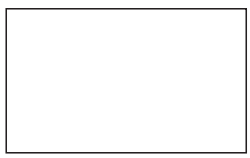


Various types of Quadrilaterals



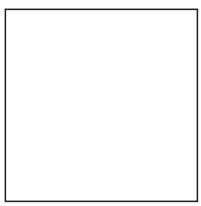
(i) Parallelogram A quadrilateral in which opposite sides are parallel is called *parallelogram*, written as $\|_{gm}$.

(ii) Rectangle A parallelogram each of whose angles is 90° is called a *rectangle*, written as rect. *ABCD*.



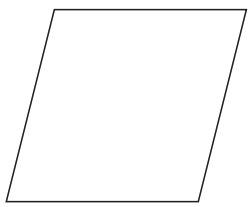
Rectangle

(iii) Square A rectangle having all sides equal is called a *square*.



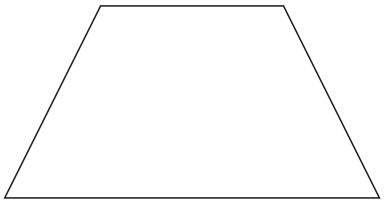
Square

(iv) Rhombus A quadrilateral having all sides equal is called a *rhombus*



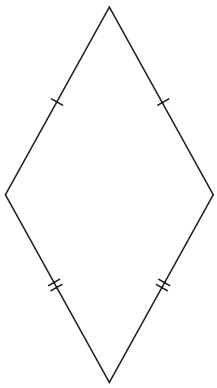
Rhombus

(v) Trapezium A quadrilateral in which two opposite sides are parallel and two opposite sides are non-parallel is called a *trapezium.*



Trapezium

(vi) Kite A quadrilateral in which pairs of adjacent sides are equal is known as *kite.*



KEY RESULTS TO REMEMBER

1. The sum of all the four angles of a quadrilateral is 360°.

2. In a parallelogram
(i) opposite sides are equal.
(ii) opposite angles are equal.
(iii) each diagonal bisects the parallelogram.
(iv) the diagonal bisect each other.

3. A quadrilateral is a ||gm
(i) if both pairs of opposite sides are equal.
or (ii) if both pairs of opposite angles are equal.
or (iii) if the diagonals bisect each other.
or (iv) if a pair of opposite sides are equal and parallel.

4. The diagonals of a rectangle are equal.

5. If the diagonals of a ||gm are equal, it is a rectangle.

6. Diagonals of a rhombus are perpendicular to each other.

7. Diagonals of a square are equal and perpendicular to each other.

8. The figure formed by joining the mid-points of the pairs of consecutive sides of a quadrilateral is a ||gm.

9. The quadrilateral formed by joining the mid-points of the consecutive sides of a rectangle is a rhombus.

10. The quadrilateral formed by joining the mid-points of the consecutive sides of a rhombus is a rectangle.

11. If the diagonals of a quadrilateral are perpendicular to each other, then the quadrilateral formed by joining the mid-points of its sides, is a rectangle.

12. The quadrilateral formed by joining the mid-points of the sides of a square, is also a square.

SECTION 4 POLYGONS

Polygon A closed plane figure bounded by line segments is called a *polygon.*

The line segments are called its *sides* and the points of intersection of consecutive sides are called its *vertices*. An angle formed by two consecutive sides of a polygon is called an *interior angle* or simply an *angle* of the polygon.

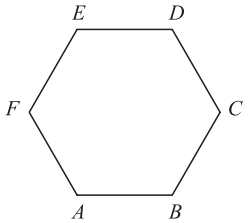
No. of sides	*Name*
3	Triangle
4	Quadrilateral
5	Pentagon
6	Hexagon
7	Heptagon
8	Octagon
9	Nonagon
10	Decagon

A polygon is named according to the number of sides, it has.

In general, a polygon of n sides is called *n-gon.* Thus, a polygon having 18 sides is called 18-gon.

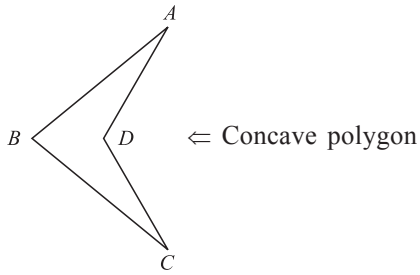
Diagonal of a Polygon Line segment joining any two non-consecutive vertices of a polygon is called its *diagonal*.

Convex Polygon If all the (interior) angles of a polygon are less than 180°, it is called a *convex polygon*. In the figure given below, *ABCDEF* is a convex polygon. In fact, it is a convex hexagon.

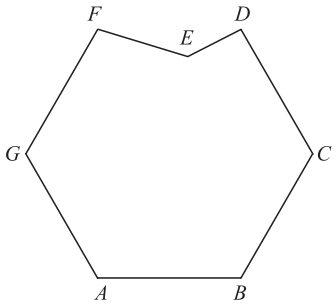


(In other words, a polygon is a convex polygon if the line segment joining any two points inside it lies completely inside the polygon).

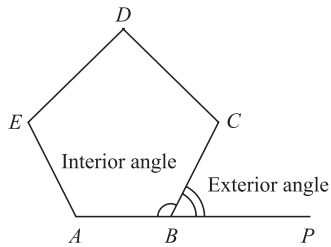
Concave Polygon If one or more of the (interior) angles of a polygon is greater than 180° i.e., reflex, it is called



concave (or re-entrant) *polygon* In the figure given below, *ABCDEFG* is a concave polygon. In fact, it is a concave heptagon.



Exterior Angle of Convex Polygon If we produce a side of polygon, the angle it makes with the next side is called an *exterior angle*. In the diagram given below, *ABCDE* is a pentagon. Its side *AB* has been produced to *P*, then $\angle CBP$ is an exterior angle.



Notes

Corresponding to each interior angle, there is an exterior angle. Also, as an exterior angle and its adjacent interior angle make a straight line, we have **an exterior angle + adjacent interior angle = 180°**

Regular Polygon A polygon is called regular polygon if all of its sides have equal length and all its angles have equal size.

Thus, in a regular polygon

(i) all sides are equal in length.

(ii) all interior angles are equal in size.

(iii) all exterior angles are equal size.

Notes

All regular polygons are convex.

KEY RESULTS TO REMEMBER

1. (a) If there is a polygon of n sides ($n \geq 3$), we can cut it into $(n - 2)$ triangles with a common vertex and so *the sum of the interior angles of a polygon of n sides would be*

$$(n - 2) \times 180° = (n - 2) \times 2 \text{ right angles}$$
$$= (2n - 4) \text{ right angles}$$

(b) If there is a regular polygon of n sides ($n \geq 3$), then its each interior angle is equal to

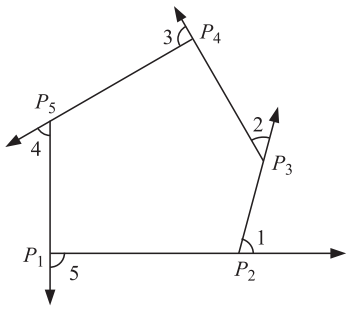
$$\left(\frac{2n-4}{n} \times 90\right)$$

(c) Each exterior angle of a regular polygon of n sides is equal to

$$= \left(\frac{360}{n}\right)^{\circ}$$

2. The sum of all the exterior angles formed by producing the sides of a convex polygon in the same order is equal to four right angles.

Explanation If in a convex polygon $P_1P_2P_3P_4P_5$, all the sides are produced in order, forming exterior angles $\angle 1$, $\angle 2$, $\angle 3$, $\angle 4$ and $\angle 5$, then $\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 = 4$ right angles.



3. If each exterior angle of a regular polygon is $x°$, then the number of sides in the polygon $= \dfrac{360°}{x}$.

Notes

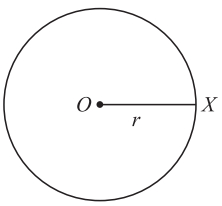
Greater the number of sides in a regular polygon, greater is the value of its each interior angle and smaller is the value of each exterior angle.

4. If a polygon has n sides, then the number of diagonals of the polygon

$$= \frac{n(n-1)}{2} - n.$$

SECTION 5 CIRCLES AND TANGENTS

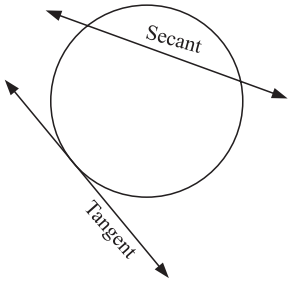
Circle A circle is a set of all those points in a plane, each one of which is at given constant distance from a given fixed point in the plane.



The fixed point is called the *centre* and the given constant distance is called the *radius* of the circle.

A circle with centre O and radius r is usually denoted by $C(O, r)$.

Tangent A line meeting a circle in only one point is called a *tangent* to the circle. The point at which the tangent line meets the circle is called the *point of contact*.

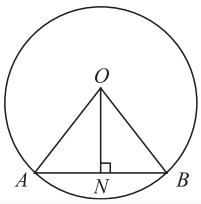


Secant A line which intersects a circle in two distinct points is called a *secant line*.

KEY RESULTS TO REMEMBER

1. The perpendicular from the centre of a circle to a chord bisects the chord.

Explanation If $ON \perp AB$, then $AN = NB$.

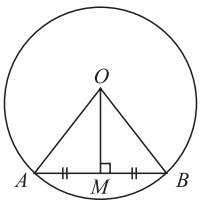


Notes

The converse of above theorem is true and can be stated as point 2.

2. The line joining the centre of a circle to the midpoint of a chord is perpendicular to the chord.

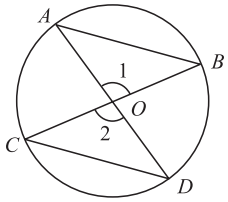
Explanation If $AM = MB$, then $OM \perp AB$.



Cor. The perpendicular bisectors of two chords of a circle intersect at its centre.

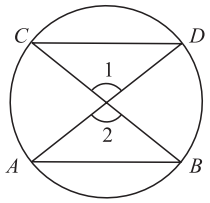
3. Equal chords of a circle subtend equal angles at the centre.

Explanation If $AB = CD$, then $\angle 1 = \angle 2$



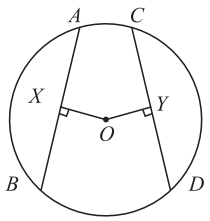
4. (Converse of above theorem) If the angles subtended by two chords at the centre of a circle are equal then the chords are equal.

Explanation If $\angle 1 = \angle 2$, then $AB = CD$



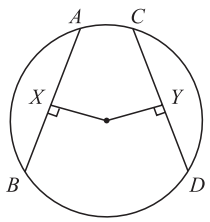
5. Equal chords of a circle are equidistant from the centre.

Explanation If the chords AB and CD of a circle are equal and if $OX \perp AB$ and $OY \perp CD$ then $OX = OY$.



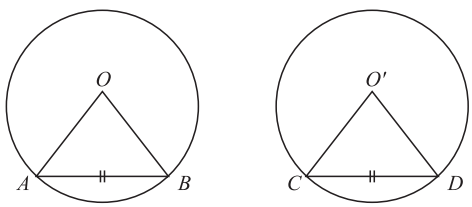
6. (Converse above theorem) Chords equidistant from the centre of the circle are equal.

Explanation If $OX \perp AB$ $OY \perp CD$ and $OX = OY$, then chords $AB = CD$



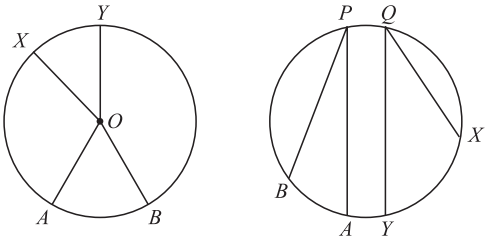
7. In equal circles (or in the same circle), equal chords cut of equal arcs.

Explanation If the chords $AB = CD$, then arc $AB =$ are CD.



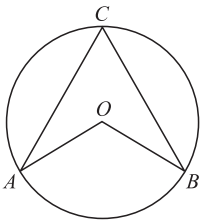
8. In equal circles (or in the same circle) if two arcs subtend equal angles at the centre (or at the circumference), the arcs are equal.

Explanation If $\angle BOA = \angle XOY$, then arc AB = arc XY or if $\angle BPA = \angle XQY$, then arc AB = arc XY.



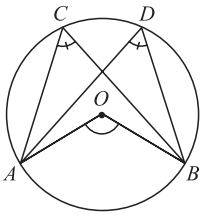
9. The angle subtended by an arc of a circle at the centre is double the angle subtended by it at any point on the remaining part of the circle.
(The theorem is popularly known as Degree Measure Theorem).

Explanation A circle, centre O, with $\angle AOB$ at the centre, $\angle ACB$ at the circumference, standing on the same arc AB, then $\angle AOB = 2\angle ACB$

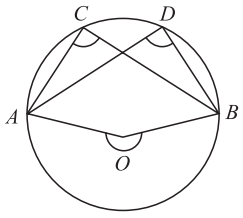


10. Angles in the same segment of a circle are equal.

Explanation A circle, centre O, $\angle ACB$ and $\angle ADB$ are angles at the circumference, standing on the same arc, then

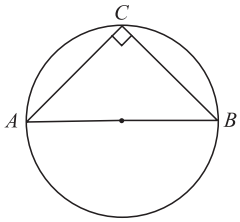


$\angle ACB = \angle ADB$
(angles in same arc)
or
(angles in same segment)



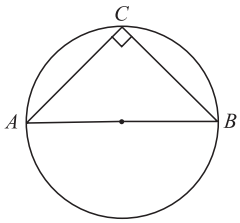
11. The angle in a semicircle is a right angle.

Explanation In the figure given below $\angle ACB = 90°$

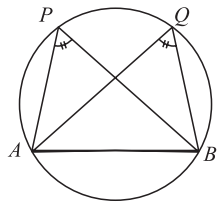


12. (Converse of above theorem) The circle drawn with hypotenuse of a right triangle as diameter passes through its opposite vertex.

Explanation The circle drawn with the hypotenuse AB of a right triangle ACB as diameter passes through its opposite vertex C.

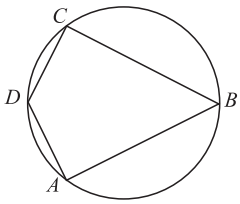


13. If $\angle APB = \angle AQB$, and if P, Q are on the same side of AB, then A, B, Q, P are concyclic i.e., lie on the same circle.



14. The sum of the either pair of the opposite angles of a cyclic quadrilateral is 180°.

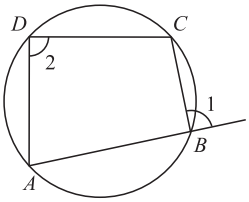
Explanation If $ABCD$ is a cyclic quadrilateral, then $\angle A + \angle C = \angle B + \angle D = 180°$



15. (Converse of above theorem) If the two angles of a pair of opposite angles of a quadrilateral are supplementary then the quadrilateral is cyclic.

16. If a side of a cyclic quadrilateral is produced then the exterior angle is equal to the interior opposite angle.

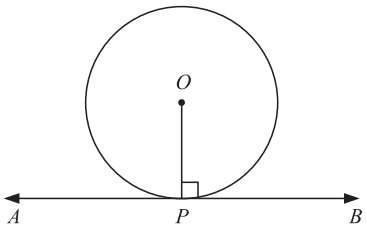
Explanation If the side AB of a cyclic quadrilateral $ABCD$ is produced then $\angle 1 = \angle 2$.



THEOREMS ON TANGENTS

17. A tangent at any point of a circle is perpendicular to the radius through the point of contact.

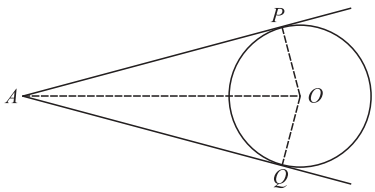
Explanation If AB is a tangent at a point P to a circle $C\ (O, r)$ then $PO \perp AB$



18. (Converse of above theorem) A line drawn through the end of a radius and perpendicular to it, is a tangent to the circle.

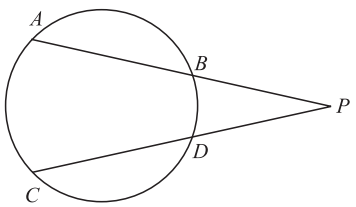
19. The lengths of two tangents drawn from an external point to a circle are equal.

Explanation If two tangents AP and AQ are drawn from a point A to a circle $C(O, r)$, then $AP = AQ$

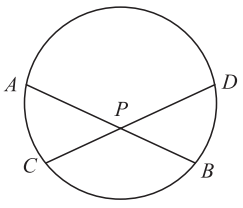


20. If two chords AB and CD intersect internally (ii) or externally (i) at a point P then

$$PA \times PB = PC \times PD$$



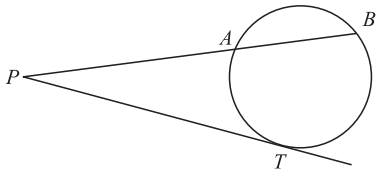
(i)



(ii)

21. If PAB is a secant to a circle intersecting the circle at A and B is a tangent segment then $PA \times PB = PT^2$ (refer the figure below).

(popularly known as Tangent-Secant theorem)



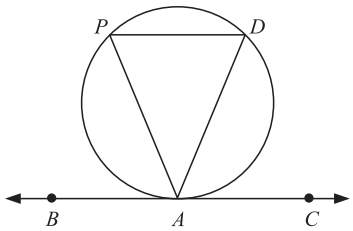
22. Alternate Segment Theorem:

In the figure below, if BAC is the tangent at A to a circle and if AD is any chord, then

$$\angle DAC = \angle APD \text{ and}$$

$$\angle PAB = \angle PDA$$

(Angles in alternate segment)

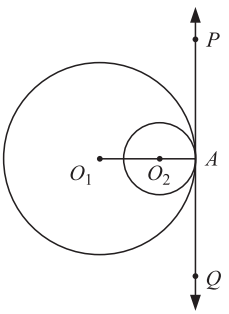


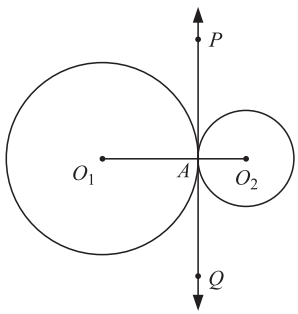
Notes

The converse of the above theorem is true.

23. If two circles touch each other internally or externally, the point of contact lies on the line joining their centres.

Explanation If two circles with centre O_1 and O_2 which touch each other internally (i) or externally (ii), at a point A then the point A lies on the line $O_1 O_2$, i.e., three points A, O_1 and O_2 are collinear.



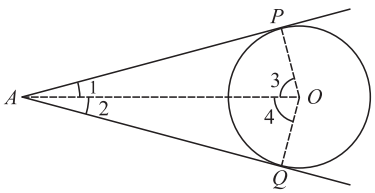


SOME USEFUL RESULTS

1. There is one and only one circle passing through three non-collinear points.
2. Two circles are congruent if and only if they have equal radii.
3. Of any two chords of a circle, the one which is greater is nearer to the centre.
4. Of any two chords of a circle, the one which is nearer to the centre is greater.
5. If two circles intersect in two points, then the line through the centres is the perpendicular bisector of the common chord.
6. Angle in a major segment of a circle is acute and angle in a minor segment is obtuse.
7. If two tangents are drawn to a circle from an external point then
   (i) they subtend equal angles at the centre.
   (ii) they are equally inclined to the segment, joining the centre to that point.

Explanation In a circle $C(O,r)$, A is a point outside it and AP and AQ are the tangents drawn to the circle

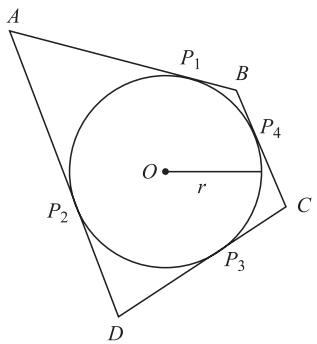
Then, $\angle 1 = \angle 2$ and $\angle 3 = \angle 4$



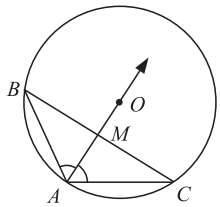
24. If a circle touches all the four sides of a quadrilateral then the sum of opposite pair of sides are equal.

Explanation If $ABCD$ is a circumscribed quadrilateral.

Then, $AB + CD = AD + BC$

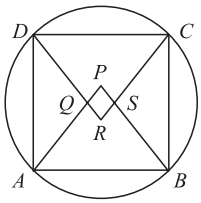


25. If two chords AB and AC of a circle are equal, then the bisector of $\angle BAC$ passes through the centre O of the circle.



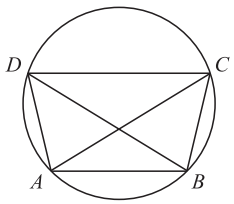
26. The quadrilateral formed by angle bisectors of a cyclic quadrilateral is also cyclic.

Explanation If $ABCD$ is a cyclic quadrilateral in which AP, BP, CR and DR are the bisectors of $\angle A$, $\angle B$, $\angle C$ and $\angle D$, respectively, then quadrilateral $PQRS$ is also cyclic.



27. A cyclic trapezium is isoceles and its diagonals are equal.

Explanation If ABC cyclic trapezium such that $AB \parallel DC$, then $AD = BC$ and $AC = BD$



28. If two opposite sides of a cyclic quadrilateral are equal, then the other two sides are parallel.

Explanation A cyclic quadrilateral $ABCD$ in which $AD = BC$

Then, $AB \parallel CD$

